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In [1]: # Program 12: Linear Regression on the Diabetes dataset
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

FILE = r"D:\Datascience-MachineLearning\Datasets\diabetes.csv" # <-- change to your file path

df = pd.read_csv(FILE)
X = df.iloc[:, :-1] # all columns except the last = features
y = df.iloc[:, -1] # last column = target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

pred = model.predict(X_test)
print("Mean Squared Error:", mean_squared_error(y_test, pred))
print("R2 score          :", r2_score(y_test, pred))

# predict for new input data (here: first 3 rows of the test set as "new" data)
new_data = X_test.iloc[:3]
print("\nNew input:")
print(new_data)
print("Predicted values:", model.predict(new_data))
```

Mean Squared Error: 0.17104527280850096

R2 score : 0.2550028117674178

New input:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
668	6	98	58	33	190	34.0	
324	2	112	75	32	0	35.7	
624	2	108	64	0	0	30.8	

	DiabetesPedigreeFunction	Age
668	0.430	43
324	0.148	21
624	0.158	21

Predicted values: [0.33550028 0.23809869 0.1510522]

In []: